M.I.N.E.R.S. Rover - BTS7960 Motor Driver & 12V Battery Wiring Guide

Hardware Setup: BTS7960 (IBT-2) 43A High-Power H-Bridge, Dedicated 12V Battery, 12V DC Motors, Arduino Mega 2560

CRITICAL SAFETY RULE: COMMON GROUND

The 12V Battery Negative Terminal (B-) MUST be connected to BOTH the BTS7960 GND pin AND an Arduino Mega GND pin for a shared common ground reference.

1. Power Terminals (Heavy Duty Screw Terminals)

BTS7960 Slot Label	Connection Target	Cable Type & Details
B+	12V Battery POSITIVE (+)	Thick Power Cable (16-18 AWG)
B-	12V Battery NEGATIVE (-)	Thick Power Cable (Connect to Arduino GND too!)
M+	12V DC Motor Wire (+)	Connect to Motor Positive (+) Terminal
M-	12V DC Motor Wire (-)	Connect to Motor Negative (-) Terminal

2. Control Logic Pins (Arduino Mega 2560 mega2560_rover_master.ino)

BTS7960 Pin	Left Motor Driver Pin	Right Motor Driver Pin	Function / Wire Target
VCC	Arduino Mega 5V Pin	Arduino Mega 5V Pin	Logic 5V Power for Isolator
GND	Arduino Mega GND Pin	Arduino Mega GND Pin	Logic Common Ground
R_EN	Arduino Pin D30 (or 5V)	Arduino Pin D32 (or 5V)	Forward Enable (Tie to 5V or D30/D32)
L_EN	Arduino Pin D30 (or 5V)	Arduino Pin D32 (or 5V)	Reverse Enable (Tie to 5V or D30/D32)
RPWM	Arduino Pin D2 (PWM)	Arduino Pin D4 (PWM)	Forward Speed PWM (0-255)
LPWM	Arduino Pin D3 (PWM)	Arduino Pin D5 (PWM)	Reverse Speed PWM (0-255)
R_IS / L_IS	LEAVE UNCONNECTED	LEAVE UNCONNECTED	Current Sense (Not Required for Driving)

Motor Direction & Speed Tuning

If a motor spins backward when driving FORWARD ('CMD:F' or 'W'), simply swap its M+ and M- wires at the screw terminal block.